

DESIGN AND IMPLEMENTATION OF A DATABASE MANAGEMENT SYSTEM FOR INTERNSHIP APPLICATION AND PROGRESS MONITORING SYSTEM

IAPMS

Internship Application and Progress Monitoring System

STUDENT

Suragana Sai Dattu

ROLL NUMBER

25WU0101140

FACULTY

Prof. Sahithi

COURSE

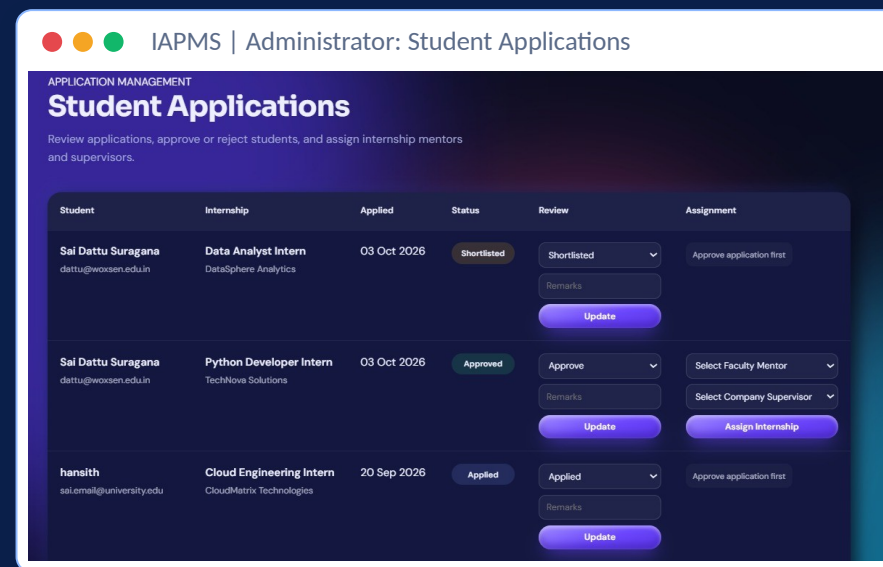
Database Management Systems

UNIVERSITY

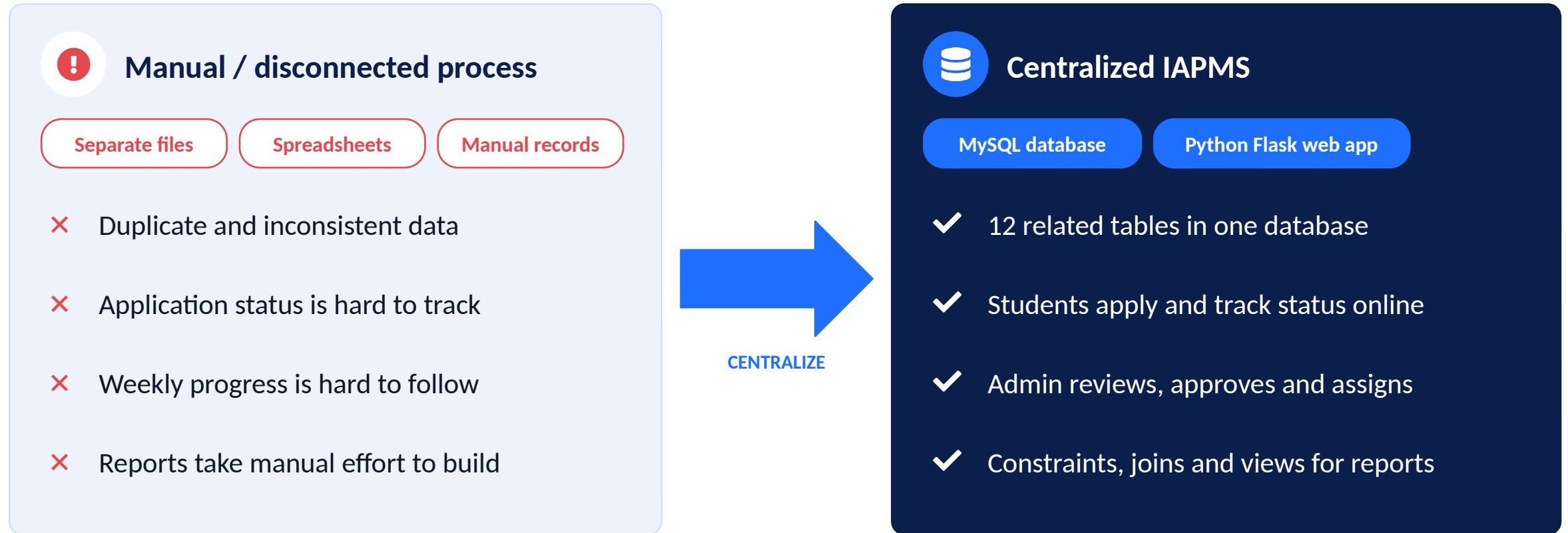
Woxsen University, School of Technology

ACADEMIC YEAR

2026-27



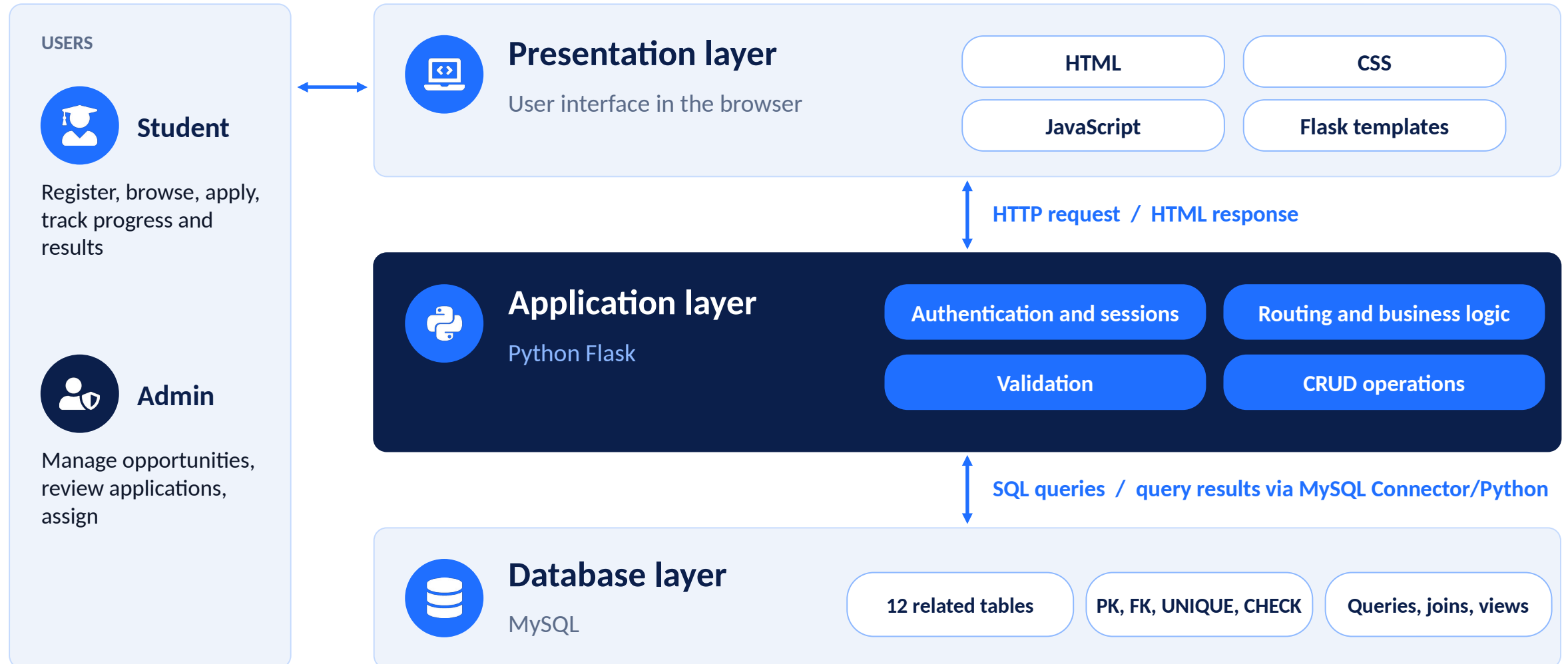
Disconnected records vs. one centralized system



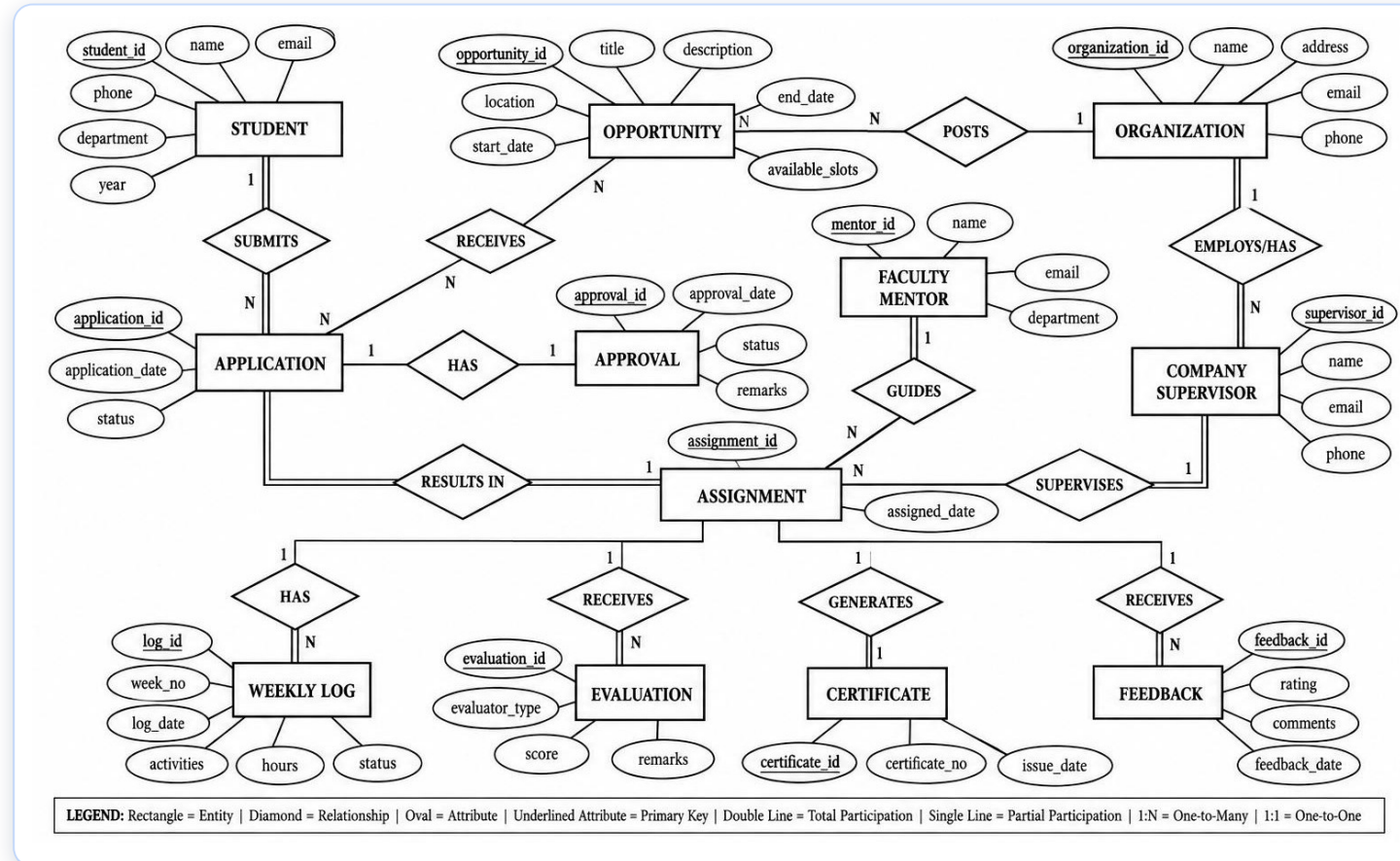
ONE CONNECTED INTERNSHIP LIFECYCLE



Three layers: Flask connects the interface to MySQL



12 entities, normalized to Third Normal Form



Final ER diagram (report, Figure 2)

PK on every table

FKs link all 12 tables

1NF → 2NF → 3NF

ENTITY GROUPS

- **Core data**
Student, Organization, Opportunity
- **Application flow**
Application, Approval
- **People and assignment**
Faculty Mentor, Company Supervisor, Assignment
- **Progress and outcomes**
Weekly Log, Evaluation, Certificate, Feedback

KEY RELATIONSHIPS

Organization	1 : N	Opportunity
Student	1 : N	Application
Application	1 : 1	Approval
Application	1 : 1	Assignment
Assignment	1 : N	Log, Evaluation, Feedback
Assignment	1 : 1	Certificate

MySQL implementation and SQL concepts

DDL

CREATE TABLE with PK, FK, UNIQUE, CHECK and DEFAULT

DML / CRUD

INSERT, SELECT, UPDATE and DELETE on internship opportunities

JOIN

Multi-table joins across student, opportunity and organization

Aggregates

COUNT applications per organization; AVG evaluation score

View

internship_assignment_report joins 7 tables

Constraints

Unique email, year 1-4, slots > 0, score 0-100, rating 1-5

REPRESENTATIVE QUERY | REPORT APPENDIX C.3

```
SELECT
    s.name AS student,
    o.title AS internship,
    org.name AS organization,
    a.application_date,
    a.status
FROM application a
JOIN student s
    ON a.student_id = s.student_id
JOIN opportunity o
    ON a.opportunity_id = o.opportunity_id
JOIN organization org
    ON o.organization_id = org.organization_id;
```

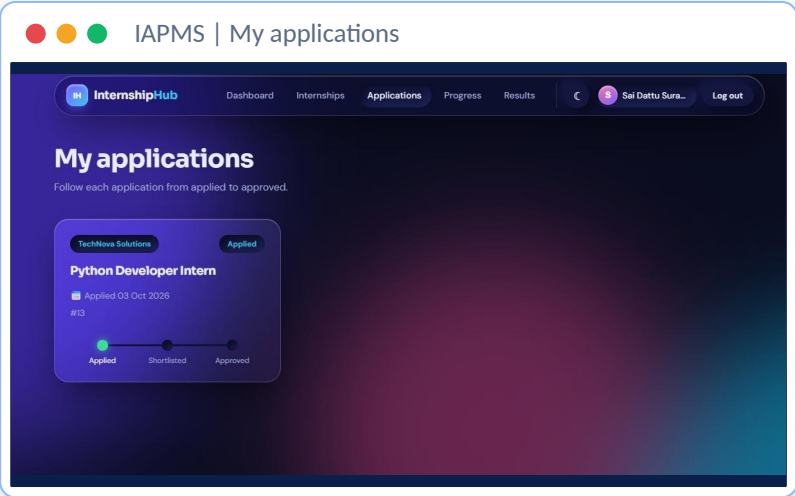
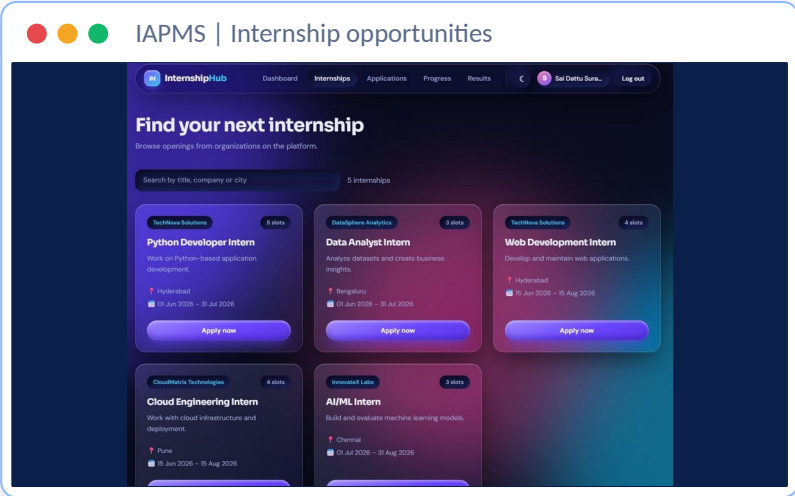
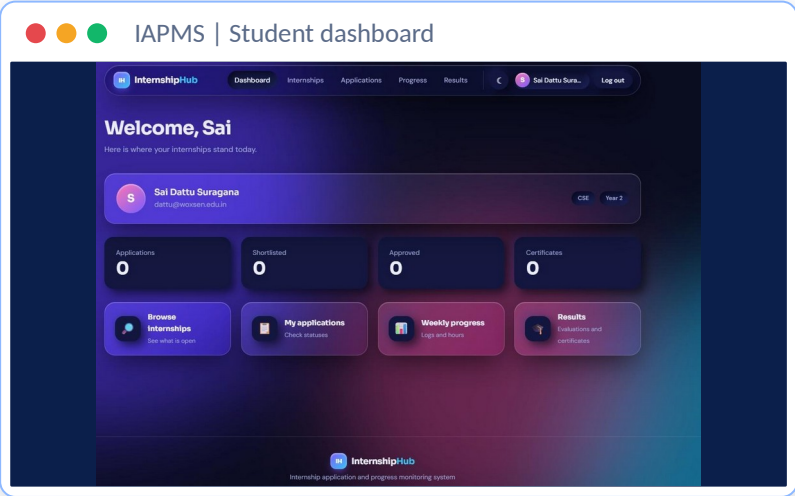
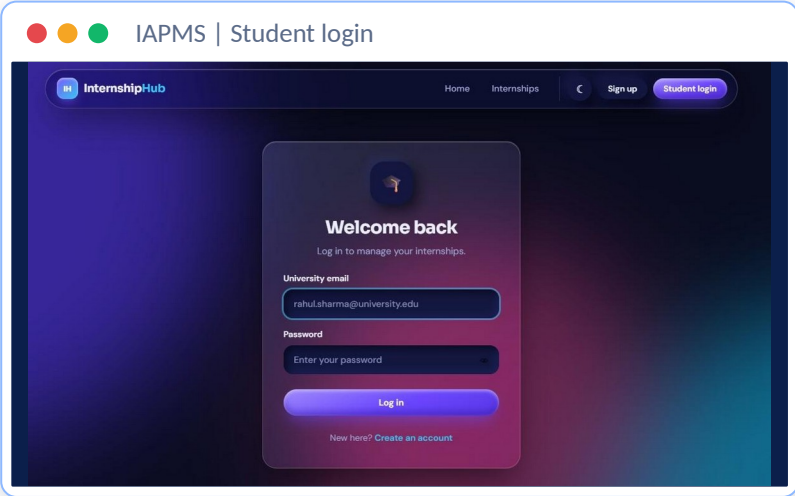
Purpose: joins student, application, opportunity and organization to list every application with its status.

student	internship	organization	application_date	status
Rahul Sharma	Python Developer Intern	TechNova Solutions	2026-05-20	Approved
Sai Dattu Suragana	Python Developer Intern	TechNova Solutions	2026-10-03	Approved
Kiran Varma	Web Development Intern	TechNova Solutions	2026-05-24	Rejected
Sai Dattu Suragana	Web Development Intern	TechNova Solutions	2026-09-20	Applied
Priya Reddy	Data Analyst Intern	DataSphere Analytics	2026-05-21	Shortlisted
Rahul Sharma	Data Analyst Intern	DataSphere Analytics	2026-05-25	Shortlisted
Sai Dattu Suragana	Data Analyst Intern	DataSphere Analytics	2026-09-20	Applied
Sai Dattu Suragana	Data Analyst Intern	DataSphere Analytics	2026-10-03	Shortlisted
Arjun Kumar	Cloud Engineering Intern	CloudMatrix Technologies	2026-05-22	Applied
hansith	Cloud Engineering Intern	CloudMatrix Technologies	2026-09-20	Applied
Sneha Rao	AI/ML Intern	InnovateX Labs	2026-05-23	Approved
Priya Reddy	AI/ML Intern	InnovateX Labs	2026-05-26	Applied

Actual output

Report Figure C.1, from the project database

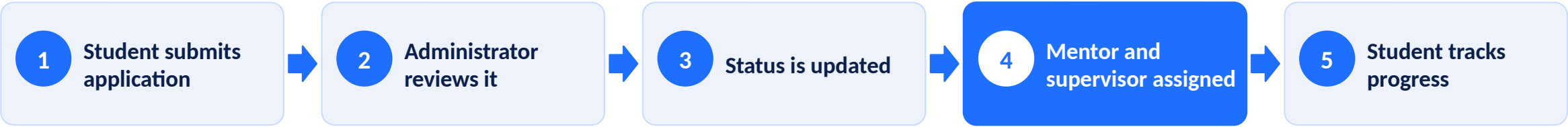
Students: from login to application tracking



STUDENT WORKFLOW

- 1 Register / Login**
Email and password; password stored hashed
- 2 Browse opportunities**
Organization, location, dates, slots
- 3 Apply**
One application per opportunity
- 4 Track status**
Applied, Shortlisted, Approved

Review, approve, then assign: two separate steps



Admin | Student Applications: review and assign

APPLICATION MANAGEMENT

Student Applications

Review applications, approve or reject students, and assign internship mentors and supervisors.

Student	Internship	Applied	Status	Review	Assignment
Sai Dattu Suragana dattu@woxsen.edu.in	Data Analyst Intern DataSphere Analytics	03 Oct 2026	Shortlisted	Shortlisted Remarks Update	Approve application first
Sai Dattu Suragana dattu@woxsen.edu.in	Python Developer Intern TechNova Solutions	03 Oct 2026	Approved	Approve Remarks Update	Select Faculty Mentor Select Company Supervisor Assign Internship
hansith sai.email@university.edu	Cloud Engineering Intern CloudMatrix Technologies	20 Sep 2026	Applied	Applied Remarks Update	Approve application first

Applied

Shortlisted

Approved

Rejected

Status values stored in APPLICATION

Admin | Dashboard

InternshipHub

Admin Dashboard Applications Add Internship C Administrator Log out

Application status updated successfully.

MANAGEMENT

Internship Opportunities

Create, update and delete internship opportunities.

+ Add Opportunity

ID	Internship	Organization	Location	Dates	Slots	Actions
#5	AI/ML Intern	InnovateX Labs	Chennai	2026-07-01 - 2026-08-31	3	Edit Delete
#4	Cloud Engineering Intern	CloudMatrix Technologies	Pune	2026-06-15 - 2026-08-15	4	Edit Delete
#3	Data Analyst Intern	DataSphere Analytics	Bengaluru	2026-06-01 - 2026-07-31	3	Edit Delete
#2	Web Development Intern	TechNova Solutions	Hyderabad	2026-06-15 - 2026-08-15	4	Edit Delete

Admin | Internship management

ID	Internship	Organization	Location	Dates	Slots	Actions
#5	AI/ML Intern	InnovateX Labs	Chennai	2026-07-01 - 2026-08-31	3	Edit Delete
#4	Cloud Engineering Intern	CloudMatrix Technologies	Pune	2026-06-15 - 2026-08-15	4	Edit Delete
#3	Data Analyst Intern	DataSphere Analytics	Bengaluru	2026-06-01 - 2026-07-31	3	Edit Delete
#2	Web Development Intern	TechNova Solutions	Hyderabad	2026-06-15 - 2026-08-15	4	Edit Delete
#1	Python Developer Intern	TechNova Solutions	Hyderabad	2026-06-01 - 2026-07-31	5	Edit Delete

IMPLEMENTED RULES

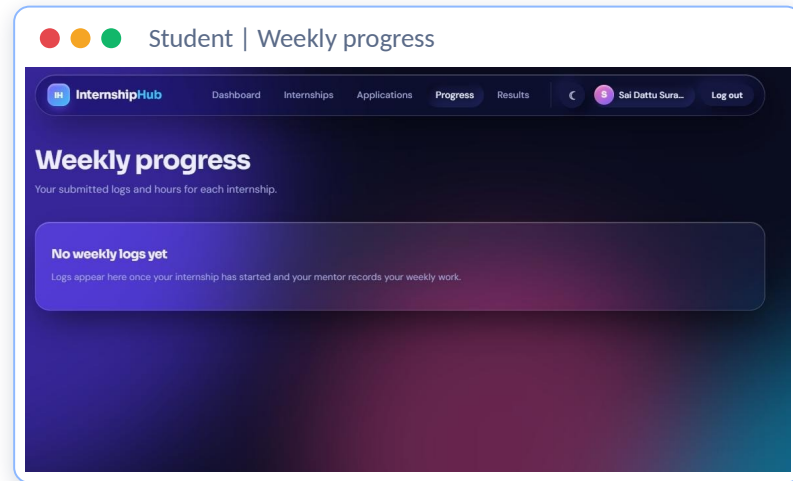
- ✓ Only approved applications can be assigned
- ✓ The faculty mentor must exist
- ✓ The supervisor must belong to the internship's organization
- ✓ One assignment per application

Approval is not assignment

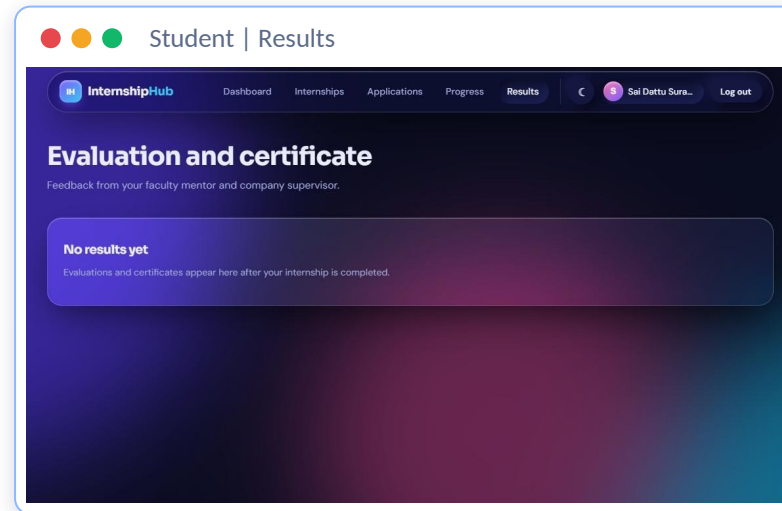
Approval: the decision and remarks (APPROVAL).

Assignment: links the approved application to a mentor and supervisor (ASSIGNMENT).

Progress monitoring, results and reporting



Empty states: logs appear once the internship starts



Empty state: results appear after the internship is completed

Weekly logs
Week no., date, activities, hours, status

Evaluation
Score (0–100), evaluator type, remarks

Certificate
Unique certificate number, issue date

Feedback
Rating (1–5), comments, date

SQL REPORTS: joins, COUNT and AVG feed the reports; one view bundles the assignment details

student

application

opportunity

organization

faculty_mentor

company_supervisor

assignment



DATABASE VIEW

internship_assignment_report

```
SELECT * FROM internship_assignment_report;
```

Mentors and supervisors are database records here; they have no separate login dashboards.

Testing, outcomes and future enhancements



Testing and outcomes

14

documented test cases, all recorded as Pass in the report

- | | |
|--|---------|
| ✓ Registration and login | T01-T04 |
| ✓ Browse internships, submit application | T05-T06 |
| ✓ Duplicate application prevented | T07 |
| ✓ Application approval and rejection | T08-T09 |
| ✓ Mentor and supervisor assignment | T10-T11 |
| ✓ Internship create, update, delete | T12 |
| ✓ Validation and error handling | T13-T14 |

Flask to MySQL connection confirmed during testing.



Limitations and future work

CURRENT LIMITATIONS

- Admin login uses project-level credentials
- No login screens for mentors or supervisors
- No email/SMS alerts or resume uploads
- Tested locally only; limited analytics

FUTURE ENHANCEMENTS

PLANNED, NOT IMPLEMENTED

- Role-based access control
- Email and SMS notifications
- Resume and document uploads
- Advanced reporting and analytics
- Cloud deployment, stronger production security

THANK YOU

Questions & Answers

IAPMS

Internship Application and Progress Monitoring System

Suragana Sai Dattu

Roll Number: 25WU0101140

